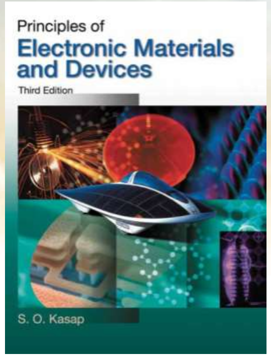
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

南方科技大学

《固态电子学》
Solid-State Electronics



Ch1-1
Elementary Materials Science Concepts

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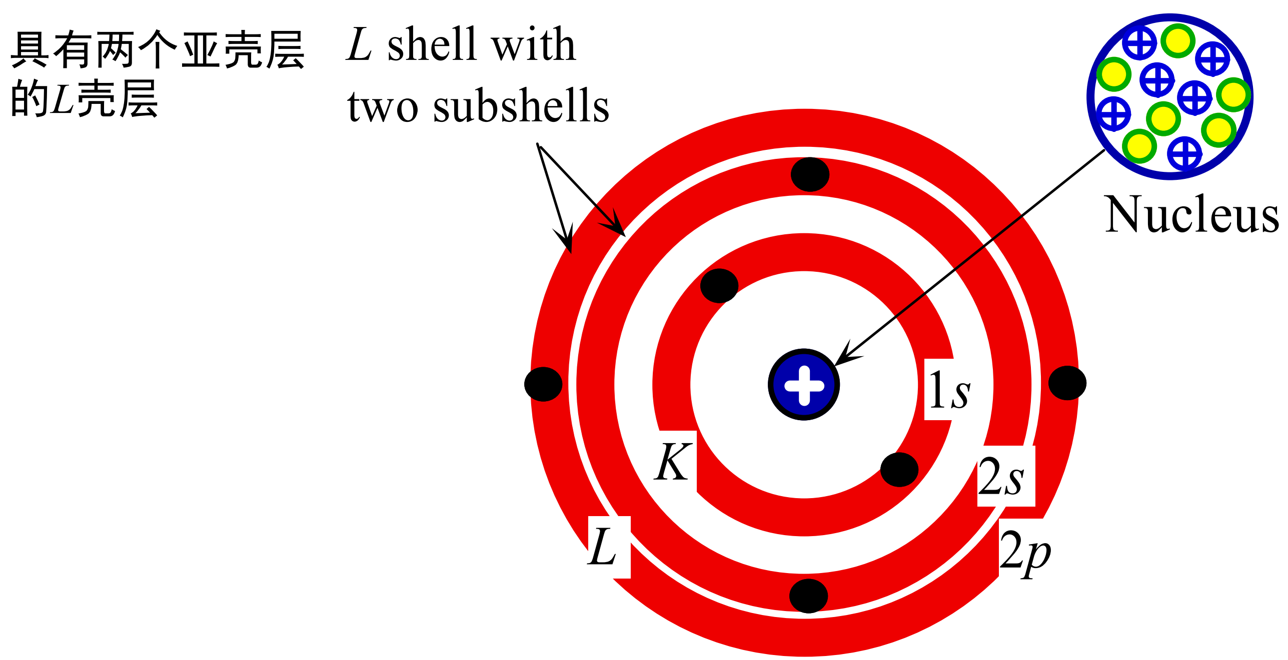
Some slides are modified from other Profs' PPTs. Their contributions are greatly acknowledged.

1.1 Atomic structure



The mass of the atom is concentrated at the nucleus（原子核）, which contains protons（质子）and neutrons（中子）

Shell model of carbon atoms $1s^2 2s^2 2p^2$ or $[He]2s^2 2p^2$

The shell model of the atom in which the electrons are confined to live within certain shells and in subshells within shells.

Table 1.1 Maximum possible number of electrons in the shells and subshells of an atom

		Subshell			
		$\ell = 0$	1	2	3
n	Shell	s	p	d	f
1	K	2			
2	L	2	6		
3	M	2	6	10	
4	N	2	6	10	14

n: shell n=1,2,3……

n=K,L,M,N………

l: sub shell l=0,1,2,3…….

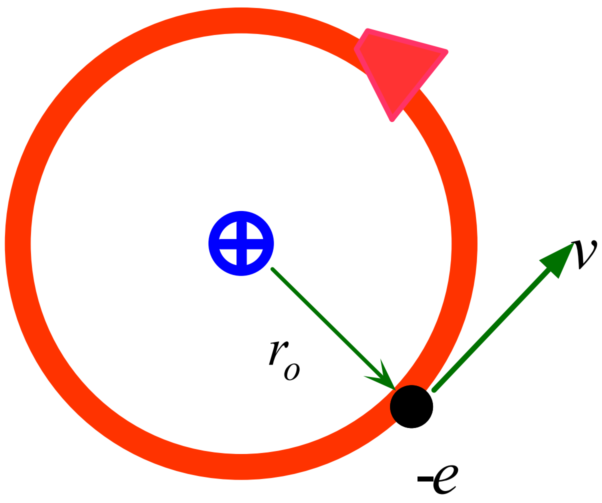
l=s,p,d,f……..

$l<n$

The number of electrons a given subshell can take is fixed by nature to be $2(2l+1)$
亚壳层中可容纳的电子数为：$2(2l+1)$

The outermost electrons are called **valence electrons**（价电子）and they determine the **valency**（化合价） of the atom.
When a subshell is full of electrons, it cannot accept any more electrons. → inert atmosphere

Stable orbit has radius r_o



The planetary model of the hydrogen atom in which the negatively charged electron orbits the positively charged nucleus.

The energy of electrons in H atoms is -13.6 eV, and the average kinetic energy is 13.6 eV.

Virial theorem （维里定理）：

总动能应该等于总（引力）势能的一半

$$\overline{E} = \overline{KE} + \overline{PE}$$

$$\overline{KE} = -\frac{1}{2}\overline{PE}$$

Kinetic energy (平均动能)：$\overline{KE}$

Potential energy (平均势能)：$\overline{PE}$

Overall energy (平均总能量)：$\overline{E}$

VIRIAL THEOREM AND THE BOHR ATOM Consider the hydrogen atom in Figure 1.2 in which the electron is in the stable 1s orbit with a radius r_o. The ionization energy of the hydrogen atom is 13.6 eV.

a. It takes 13.6 eV to ionize the hydrogen atom, i.e., to remove the electron to infinity. If the condition when the electron is far removed from the hydrogen nucleus defines the zero reference of energy, then the total energy of the electron within the H atom is -13.6 eV. Calculate the average PE and average KE of the electron.

b. Assume that the electron is in a stable orbit of radius r_o around the positive nucleus. What is the Coulombic PE of the electron? Hence, what is the radius r_o of the electron orbit?

c. What is the velocity of the electron?

d. What is the frequency of rotation (oscillation) of the electron around the nucleus?

SOLUTION

a. From Equation 1.1 we obtain

$$\overline{E} = \overline{PE} + \overline{KE} = \tfrac{1}{2}\overline{PE}$$

or

$$\overline{PE} = 2\overline{E} = 2 \times (-13.6 \text{ eV}) = -27.2 \text{ eV}$$

The average kinetic energy is

$$\overline{KE} = -\tfrac{1}{2}\overline{PE} = 13.6 \text{ eV}$$

b. The Coulombic *PE* of interaction between two charges Q_1 and Q_2 separated by a distance r_o, from elementary electrostatics, is given by

库仑作用势：
$$PE = \frac{Q_1 Q_2}{4\pi\varepsilon_o r_o} = \frac{(-e)(+e)}{4\pi\varepsilon_o r_o} = -\frac{e^2}{4\pi\varepsilon_o r_o}$$

where we substituted $Q_1 = -e$ (electron's charge), and $Q_2 = +e$ (charge of nucleus). Thus the radius r_o is

$$r_o = -\frac{(1.6\times10^{-19}\ \text{C})^2}{4\pi(8.85\times10^{-12}\ \text{F m}^{-1})(-27.2\,\text{eV}\times1.6\times10^{-19}\ \text{J/eV})}$$

$$= 5.29\times10^{-11}\ \text{m} \qquad \text{or} \qquad 0.0529\ \text{nm}$$

which is called the **Bohr radius** (also denoted a_o). 玻尔半径

c. Since $KE = \frac{1}{2}m_e v^2$, the average velocity is

$$v = \sqrt{\frac{\overline{KE}}{\frac{1}{2}m_e}} = \sqrt{\frac{13.6\ \text{eV}\times1.6\times10^{-19}\ \text{J/eV}}{\frac{1}{2}(9.1\times10^{-31}\ \text{kg})}} = 2.19\times10^6\ \text{m s}^{-1}$$

d. The period of orbital rotation T is

$$T = \frac{2\pi r_o}{v} = \frac{2\pi(0.0529\times10^{-9}\ \text{m})}{2.19\times10^6\ \text{m s}^{-1}} = 1.52\times10^{-16}\ \text{seconds}$$

The orbital frequency $\nu = 1/T = 6.59\times10^{15}\ \text{s}^{-1}$ (Hz).

1.2 Atomic Mass and Mole

Atomic Mass: 1/12 of the mass of a carbon atom. The Avogadro's constant of an atom of any substance has the same mass as the atomic mass expressed in grams.

原子质量：碳原子质量的1/12。任何物质的原子的阿伏伽德罗常数具有的质量和用克表示的原子质量相等

$$n_A = \frac{w_A/M_A}{w_A/M_A + w_B/M_B} \quad \text{and} \quad n_B = 1 - n_A$$

COMPOSITIONS IN ATOMIC AND WEIGHT PERCENTAGES Consider a Pb–Sn solder that is 38.1 wt.% Pb and 61.9 wt.% Sn (this is the eutectic composition with the lowest melting point). What are the atomic fractions of Pb and Sn in this solder?

SOLUTION

For Pb, the weight fraction and atomic mass are respectively $w_A = 0.381$ and $M_A = 207.2$ g mol^{-1} and for Sn, $w_B = 0.619$ and $M_B = 118.71$ g mol^{-1}. Thus, Equation 1.2 gives

$$n_A = \frac{w_A/M_A}{w_A/M_A + w_B/M_B} = \frac{(0.381)/(207.2)}{0.381/207.2 + 0.619/118.71}$$

$$= 0.261 \quad \text{or} \quad 26.1 \text{ at.\%}$$

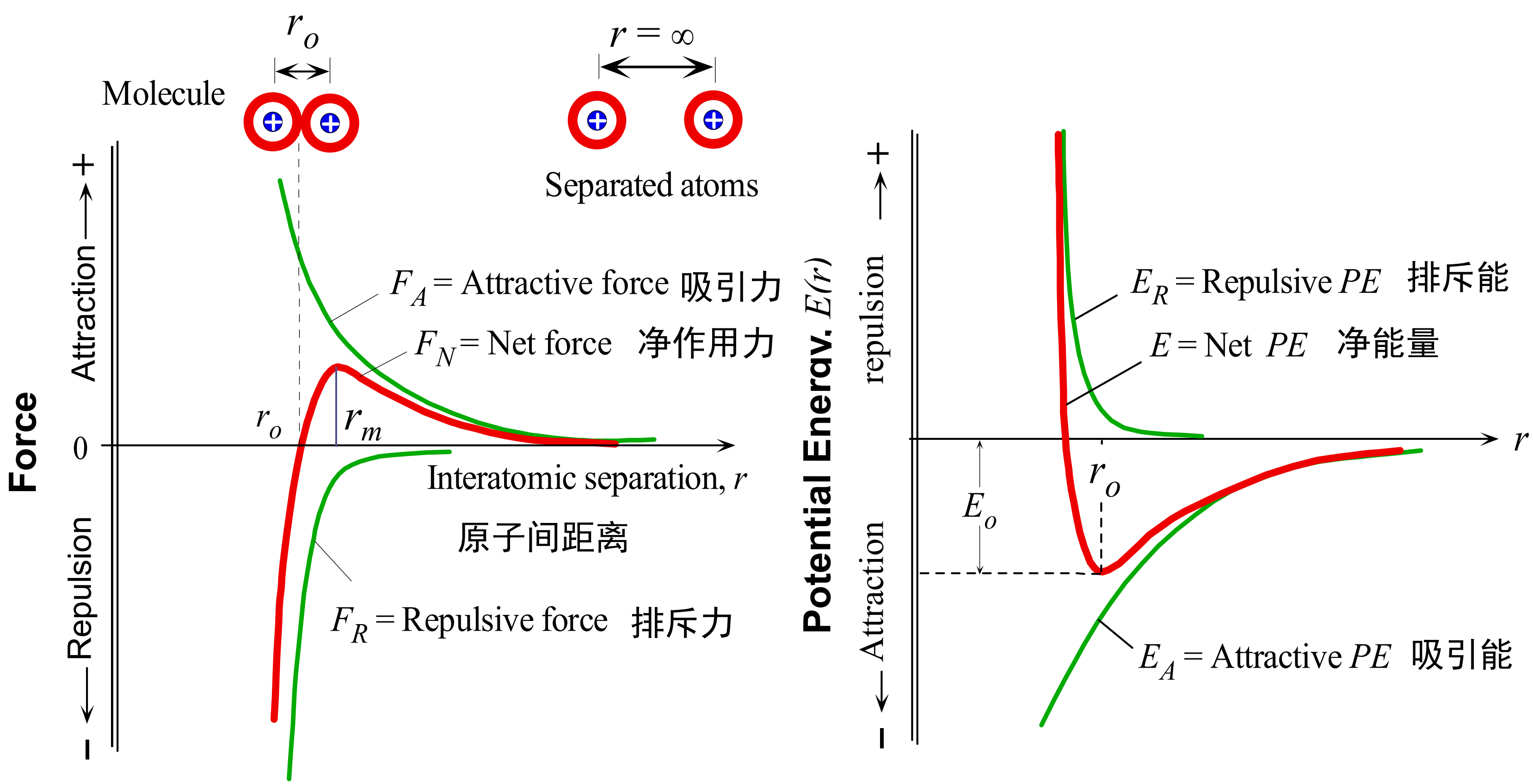
and

$$n_B = \frac{w_B/M_B}{w_A/M_A + w_B/M_B} = \frac{(0.619)/(118.71)}{0.381/207.2 + 0.619/118.71}$$

$$= 0.739 \quad \text{or} \quad 73.9 \text{ at.\%}$$

Thus the alloy is 26.1 at.% Pb and 73.9 at.% Sn which can be written as $Pb_{0.261}Sn_{0.739}$.

1.3.1 Molecules and general bonding principles

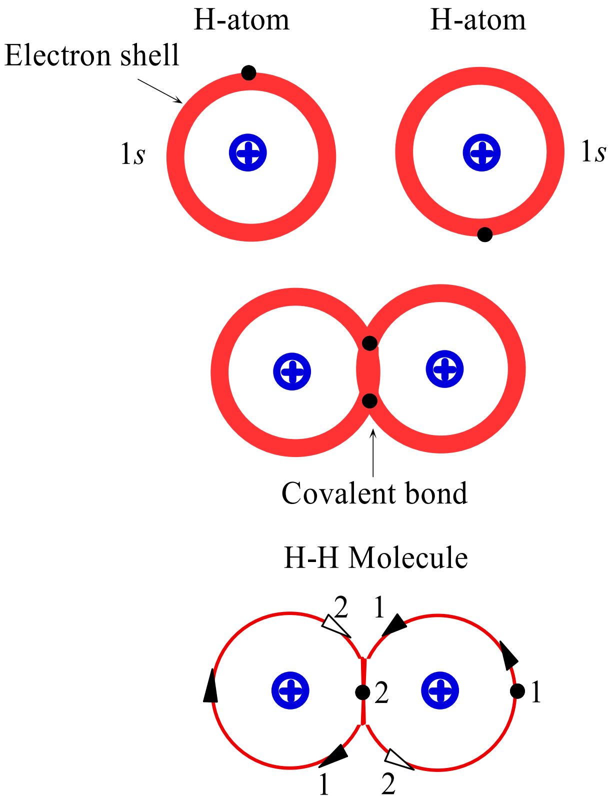


Net force between atoms
原子间的净作用力

$$F_N = F_A + F_R$$

$$F_N = \frac{dE}{dr}$$

1.3.2 Covalently bond 共价键

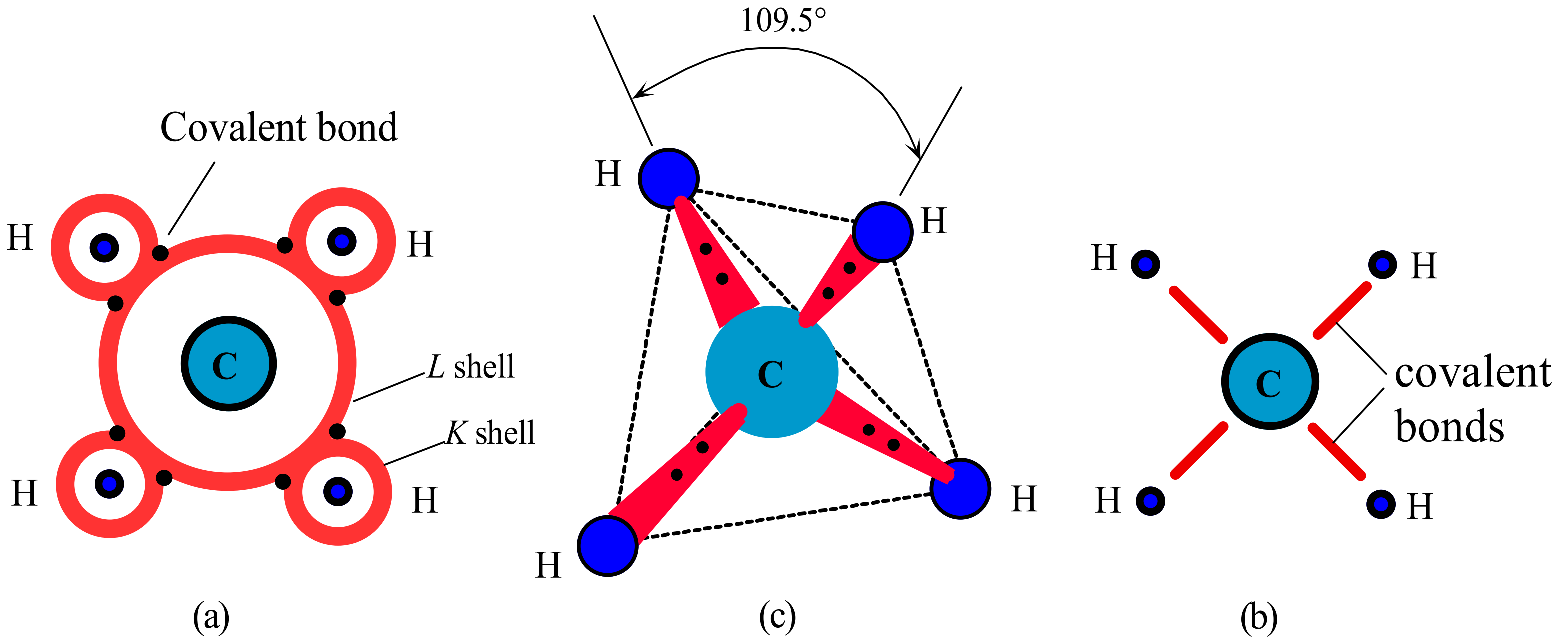


Formation of a covalent bond between two H atoms leads to the H_2 molecule. Electrons spend majority of their time between the two nuclei which results in a net attraction between the electrons and the two nuclei which is the origin of the covalent bond .

Two atoms can form a bond with each other by sharing some or all of their valence electrons and thereby reducing the overall potential energy of the combination. The covalent band results from the sharing of valence electrons to complete the subshells of each atom.

两个原子成键时共用它们的一部分价电子从而使整个结合势能降低。共价键的形成即是原子间共用电子而使每个原子的亚壳层都填满。

The two hydrogen atoms form a H_2 molecule by covalent bond, and the electrons stay longer between the two atoms.
两个氢原子之间以共价键形成H_2分子，电子在两个原子间停留时间较长。



(a) Covalent bonding in methane, CH_4, involves four hydrogen atoms sharing electrons with one carbon atom. Each covalent bond has two shared electrons. The four bonds are identical and repel each other.

(b) Schematic sketch of CH_4 on paper.

(c) In three dimensions, due to symmetry, the bonds are directed towards the corners of a tetrahedron. 四面体

Typical covalent crystals: diamond, silicon, germanium

Hardness, melting point, boiling point are very high, poor conductivity at low temperature

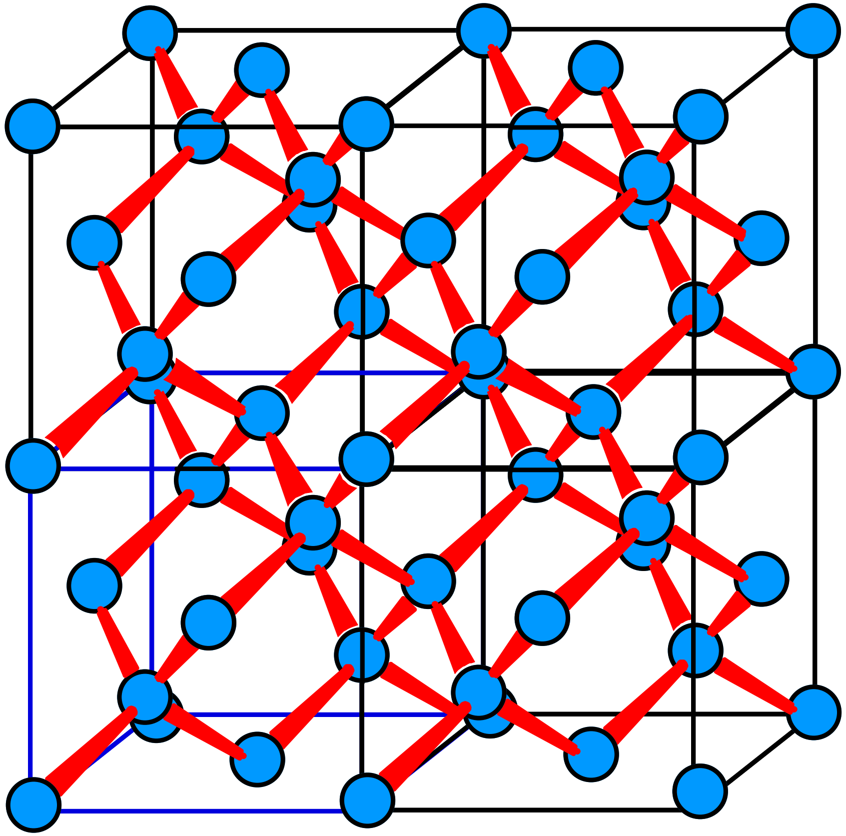
Diamond

Hardness： 10
Density： 3.5 g/cm^3
Melting point： 4000 °C，

Silicon

Hardness ： 6.5
Density ： 2.32-2.34 g/cm^3
Melting point ： 1410 °C

Crystals bonded by covalent bonding are called covalent crystals（共价晶体）, or atomic crystals（原子晶体）, or homopolar crystals（同极晶体）. They are not easy to lose the outer valence electrons, and it is not easy to obtain the valence electrons, the atoms are usually combined with each other.



Due to the strong Coulombic attraction between the shared electrons and the positive nuclei.

High melting temperature , hard solids（高熔点，高硬度）

Non-ductile, non-malleable（无塑性，延展性）
Insoluble（难溶）

Poor conductivity（电导率低）

The diamond crystal is a covalently bonded network of carbon atoms. Each carbon atom is bonded covalently to four neighbors forming a regular three dimensional pattern of atoms which constitutes the diamond crystal.

Coordination Number (配位数): the number of nearest neighbors for a given atom in the solid

Characteristics of Covalent Bond

1. **Directionality**: Atom forms a covalent bond in a particular direction.

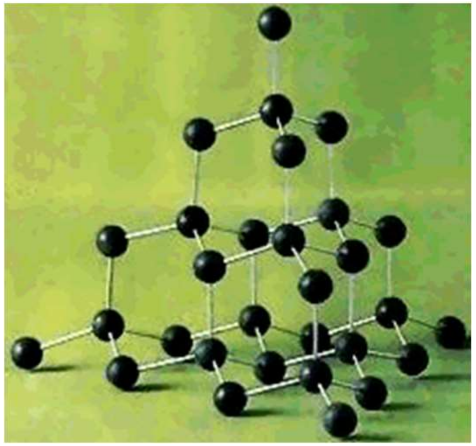
The direction of electron cloud density is large

The spatial motion of the electron pair around the two cores is most likely to occur in the space between the two cores.

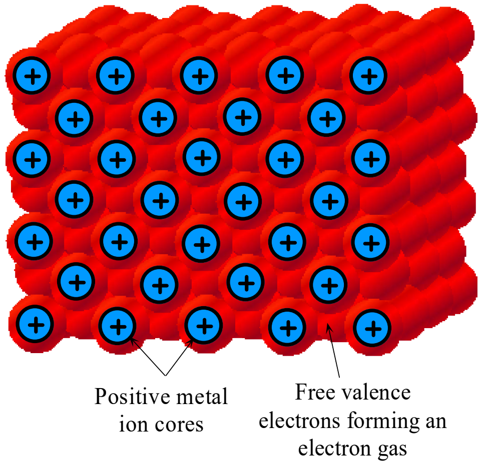
2. **Saturation**： An atom can only form a certain number of covalent bonds.

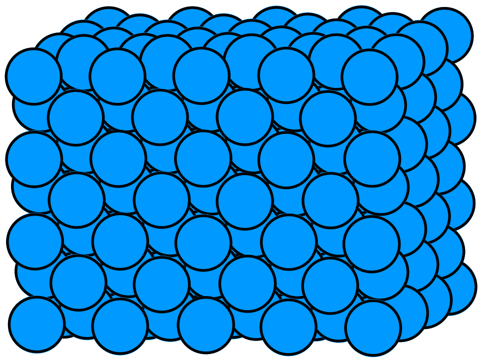
- The number of valence electrons is less than 4, and the number of covalent bonds is equal to the number of valence electrons
- The valence electron number is greater than 4, and the sum of the covalent bond and the valence electron number is equal to 8

For Diamond： Each atom has four valence electrons, forming exactly four covalent bonds with the four nearest neighbor atoms, and each pair of atoms forming a covalent bond "shares" a pair of valence electrons.



1.3.3 Metallic bond 金属键





Nondirectional nature of the bond

Ductile：延展性
High electrical conductivity，导电性好
good thermal conductivities，导热性好

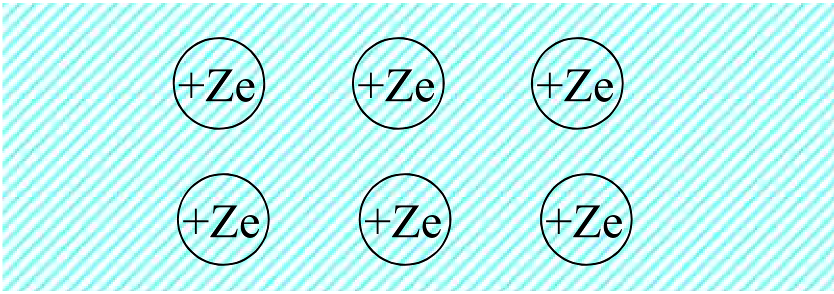
In metallic bonding the valence electrons from the metal atoms form a "cloud of electrons" which fills the space between the metal ions and "glues" the ions together through the coulombic attraction between the electron gas and positive metal ions.

在金属键中，金属原子中的价电子形成电子云，金属离子通过离子与电子之间的库仑力胶合在一起。

1. Metallic bond formation

The atom loses some or all of the valence electrons. The valence electrons are no longer bound to an atom, but are shared by all the ions. The electron cloud spreads throughout the crystal and is approximately evenly distributed to form a free electron gas.

The ions that lose the valence electrons are similar to "metal's positive ions are merged in the electronic sea".

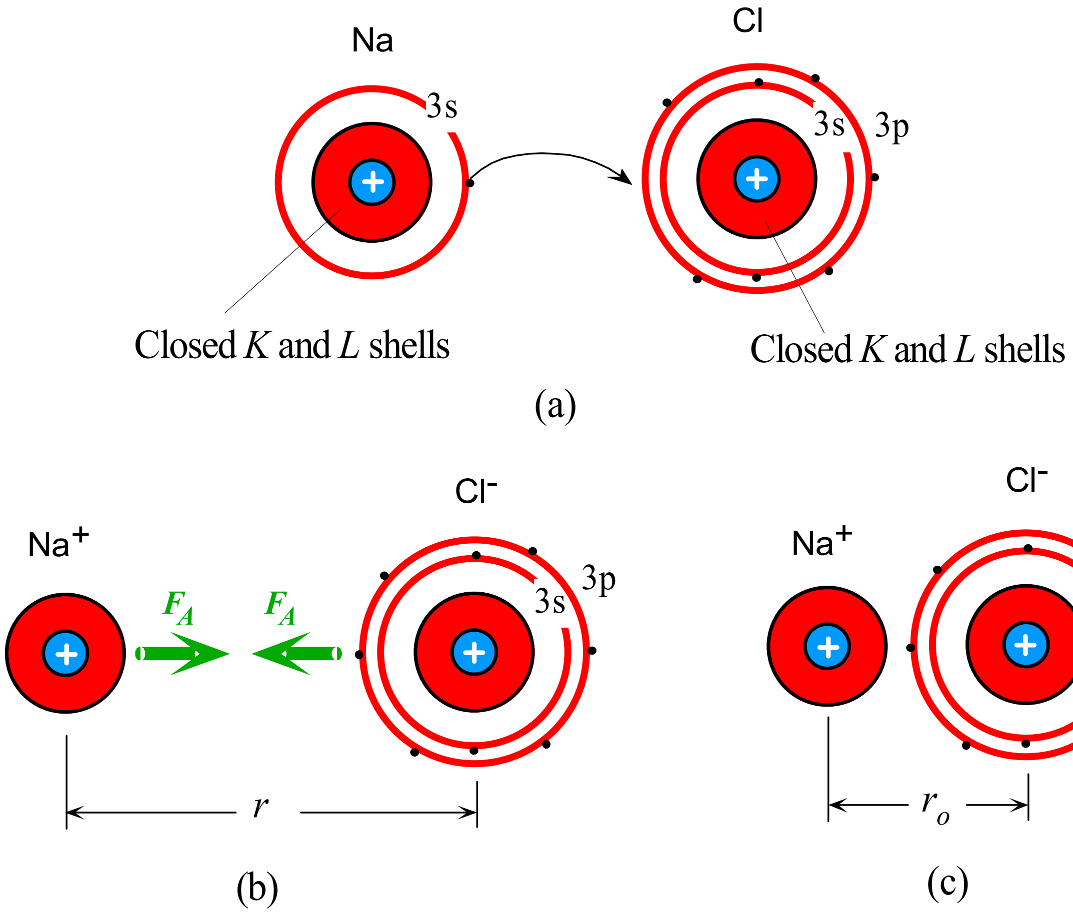


The Coulomb interaction between the free electron gas and the positive ion interact with each other to form a metal crystal.

2. Characteristics of metal crystal structure

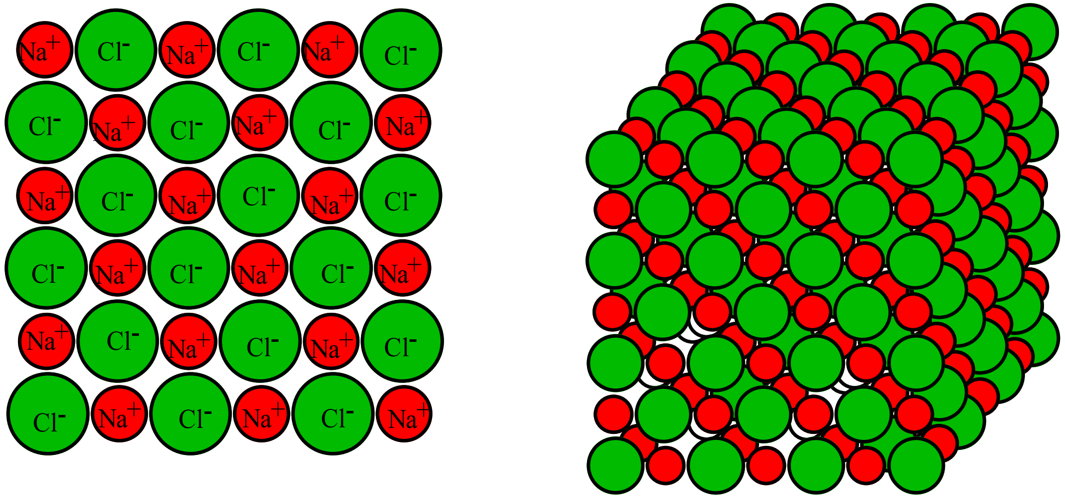
- Closed pack structure, high coordination number
- Electronic free movement
- Metallic bonds have no obvious directionality
- Ductility, good electrical and thermal conductivity

1.3.4 Ironically bond 离子键



Energy of ionization（电离能）: The energy to remove the electron from the atom.

The formation of an ionic bond between Na and Cl atoms in NaCl.
The attraction is due to coulombic forces.

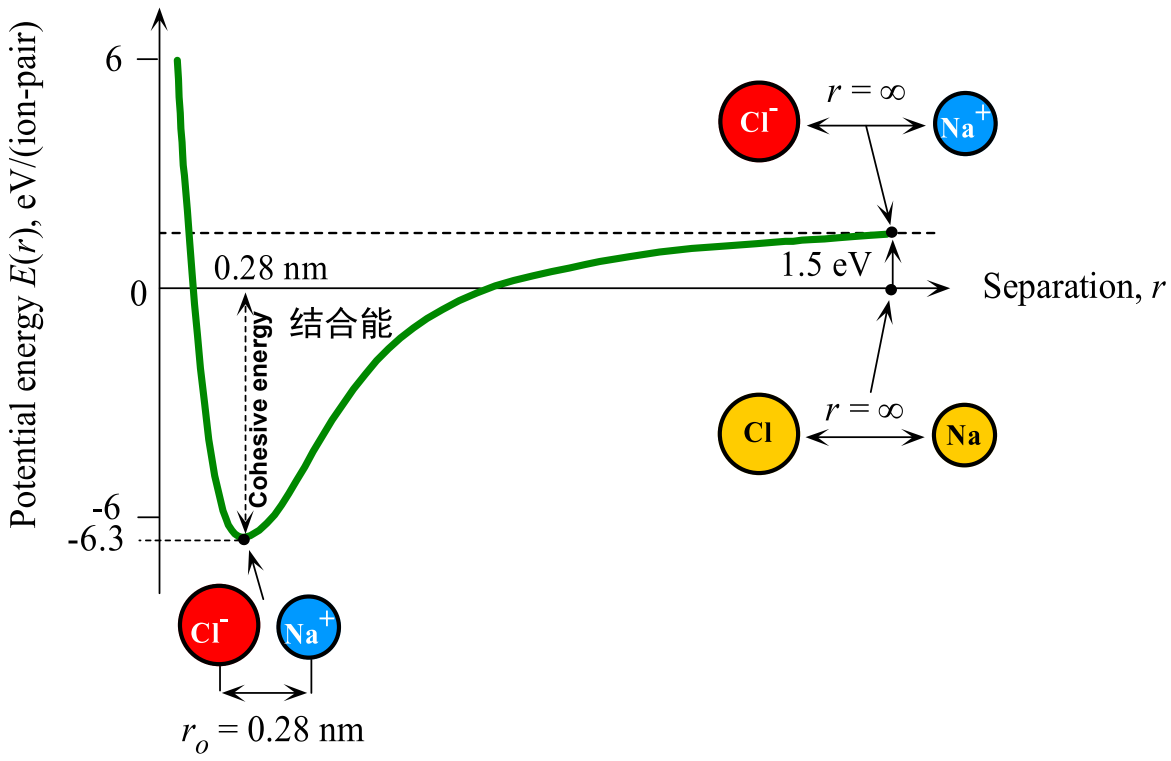


(a) (b)

(a) A schematic illustration of a cross section from solid NaCl. NaCl solid is made of Cl^- and Na^+ ions arranged alternatingly so that the oppositely charged ions are closest to each other and attract each other. There are also repulsive forces between the like-ions. In equilibrium the net force acting on any ion is zero.
(b) Solid NaCl.

NaCl的示意图，带相反电荷的离子彼此靠近且相互吸引，带同样极性电荷的离子相互排斥，在平衡时，作用于任意离子的净作用力为零。

The coordination number for both cations and anions in the NaCl crystal is 6.
配位数为6.



Sketch of the potential energy per ion-pair in solid NaCl. Zero energy corresponds to neutral Na and Cl atoms infinitely separated.

Ionic crystal: LiF, MgO, CsCl, ZnS

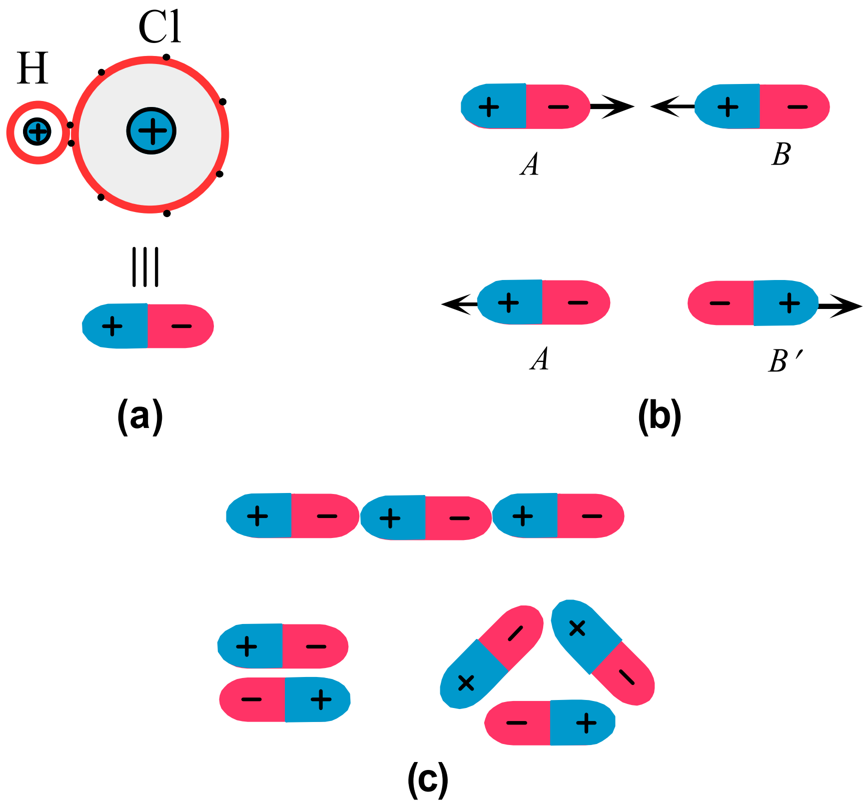
Soluble in water：一定的水溶性

Electrical insulators：绝缘

High melting temperatures：高熔点

Lower thermal conductivity：低热导率

1.3.5 Secondary bonding 弱键

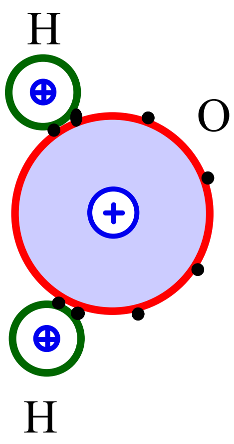


van der Waals force （范德瓦尔斯力）：electrostatic attraction between the electron distribution of one atom and the positive nucleus of the other.
一个原子的核外电子与另一个原子的原子核间的静电吸引所致。

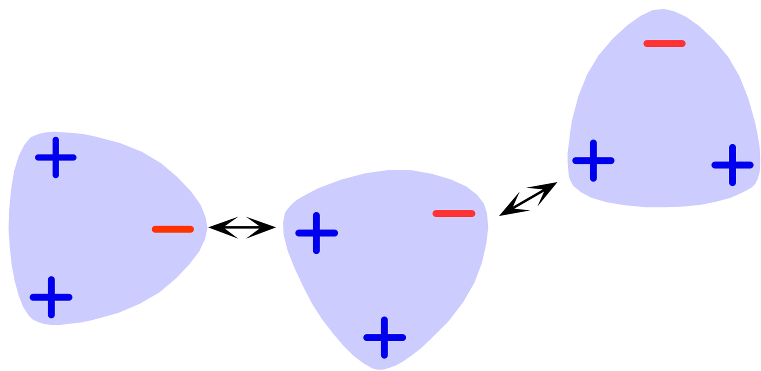
An **electric dipole moment** occurs whenever a negative and a positive charge of equal magnitude are separated by a distance.
只要出现等量的正负电荷相互分离的时刻就会产生一个点偶极子

(a) A permanently polarized molecule is called a an electric dipole moment. 被称为电偶极子的永久化分子
(b) Dipoles can attract or repel each other depending on their relative orientations. 偶极子根据其相对取向而相互排斥或吸引
(c) Suitably oriented dipoles attract each other to form van der Waals bonds. 取向合理的偶极子相互吸引形成范德瓦尔斯键

Van der Waals hydrogen bond 范德瓦尔斯氢键



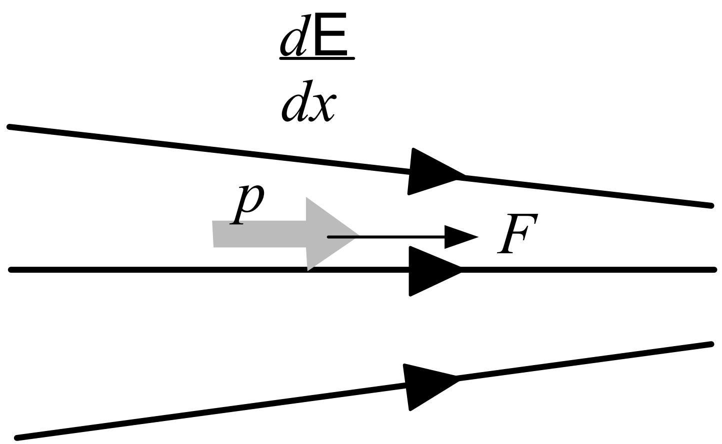
(a)



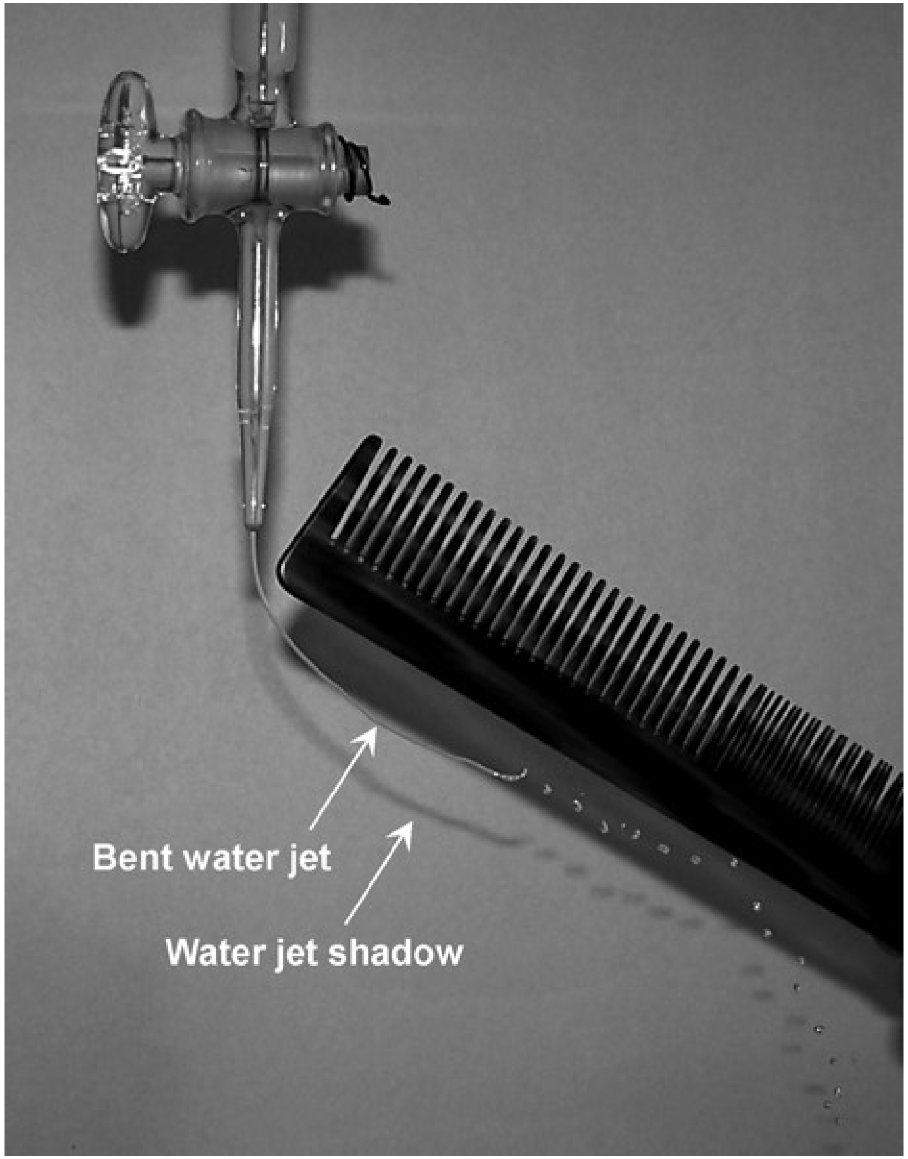
(b)

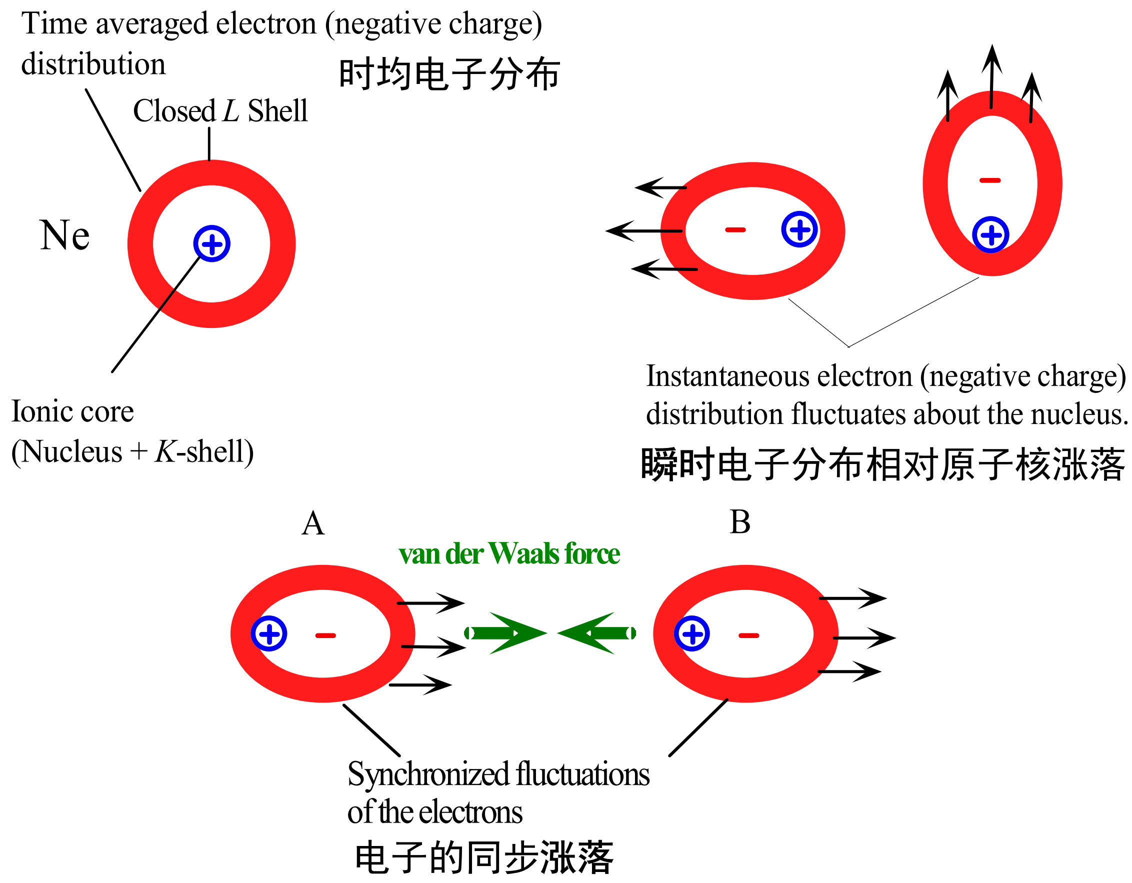
The origin of van der Waals bonding between water molecules.
(a) The H_2O molecule is polar and has a net permanent dipole moment.
(b) Attractions between the various dipole moments in water gives rise to van der Waals bonding.

(a) H_2O分子是极性分子并具有净的永久偶极矩。
(b) 水中不同偶极距间的吸引产生范德瓦尔斯键。



A dipole moment in a nonuniform field experiences a net force F that depends on the dipole moment p and the field gradient dE/dx. When a charged comb (by combing hair) is brought close to a water jet, the field from the comb attracts the polarized water molecules toward higher fields.
[See Question 7.7 in Chapter 7]





When two Ne atoms approach each other , the rapidly fluctuation negative charge distribution on one affects the motion of the negative charge distribution on the other. A lower energy configuration is produced.

两个Ne沿着相互靠近，一个原子周围负电荷分布的快速涨落影响到另一原子周围的负电荷分布，形成一个低的能量结构。

Induced dipole-induced dipole interaction and the resulting van der Waals force.
诱导偶极子与诱导偶极子之间的相互作用产生范德瓦尔斯力。

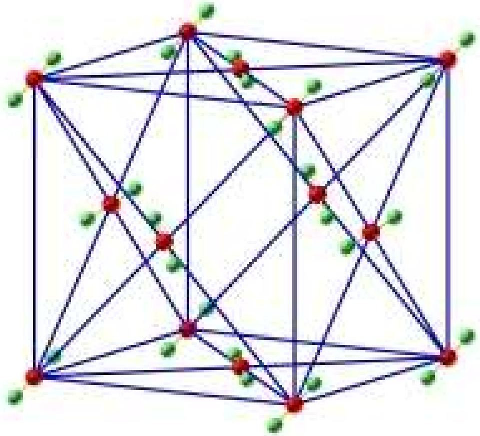
Induced-dipole-induced-dipole (诱导偶极子-诱导偶极子）：
due to synchronization of the electronic motions around the nuclei.

Typical molecular crystals: crystals formed by most gases at low temperatures
most organic compound crystal

Low melting point and binding energy, low hardness

dry ice

Density： 1.560g/cm^3 (-78°C)
Melting point： -57°C
Freezing point： -78.5°C



dry ice and its crystal structure

Hybrid Bonds →

- When covalent bonds exist between atoms of different classes (such as GaAs), the shared electrons are not equally shared, because the two ion cores have different ability to attract electrons. Gradually have the characteristics of ionic bonds.
- The ionic bond characteristic in the covalent bond is called a polar bond.

ENERGY OF SECONDARY BONDING Consider the van der Waals bonding in solid argon. The potential energy as a function of interatomic separation can generally be modeled by the **Lennard–Jones 6–12 potential energy curve,** that is,

雷纳德–琼斯势能关系

$$E(r) = -Ar^{-6} + Br^{-12}$$

where A and B are constants. Given that $A = 8.0 \times 10^{-77}$ J m^6 and $B = 1.12 \times 10^{-133}$ J m^{12}, calculate the bond length and bond energy (in eV) for solid argon.

SOLUTION

Bonding occurs when the potential energy is at a minimum. We therefore differentiate the Lennard–Jones potential $E(r)$ and set it to zero at $r = r_o$, the interatomic equilibrium separation or

$$\frac{dE}{dr} = 6Ar^{-7} - 12Br^{-13} = 0 \qquad \text{at } r = r_o$$

$$r_o = \left[\frac{2B}{A}\right]^{1/6}$$

Substituting $A = 8.0 \times 10^{-77}$ and $B = 1.12 \times 10^{-133}$ and solving for r_o, we find

$$r_o = 3.75 \times 10^{-10} \text{ m} \qquad \text{or} \qquad 0.375 \text{ nm}$$

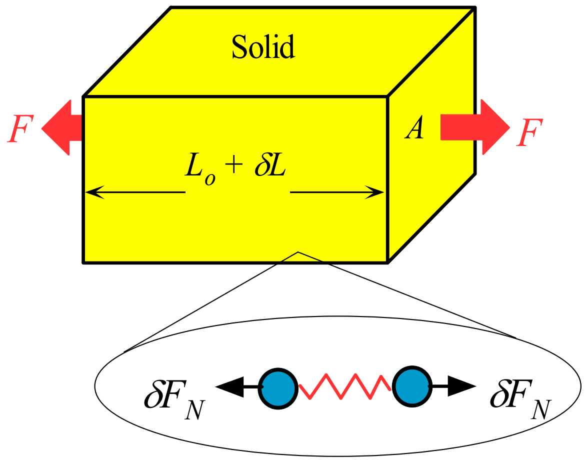
When $r = r_o = 3.75 \times 10^{-10}$ m, the potential energy is at a minimum and corresponds to $-E_{\text{bond}}$, so

$$E_{\text{bond}} = \left|-Ar_o^{-6} + Br_o^{-12}\right| = \left|-\frac{8.0 \times 10^{-77}}{(3.75 \times 10^{-10})^6} + \frac{1.12 \times 10^{-133}}{(3.75 \times 10^{-10})^{12}}\right|$$

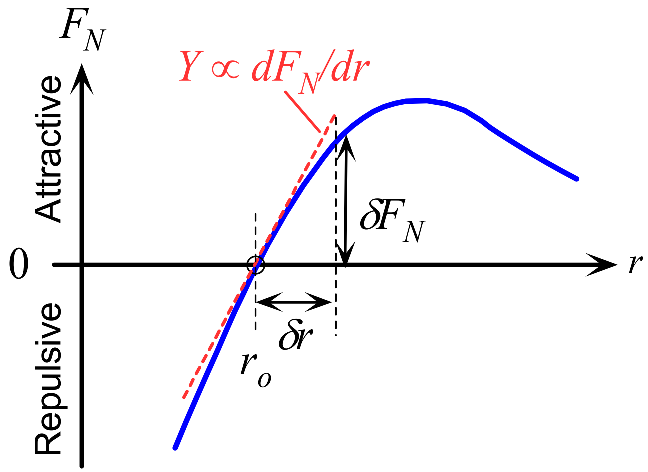
that is,

$$E_{\text{bond}} = 1.43 \times 10^{-20} \text{ J} \qquad \text{or} \qquad 0.089 \text{ eV}$$

Elastic modulus, Young's Modulus Y，弹性模量



(a)



(b)

Thomas Young

(a) Applied forces F strech the solid elastically from L_o to δL . The force is divided amongst chains of atoms that make the solid. Each chain carriers a force δF_N.

(b) In equilibrium, the applied force is balanced by the net force δF_N between the atoms as a result of their increased separation.

A stress σ （应力）： defined as the force per unit area F/A

The strain ε（应变）： the fractional increase in the length of the solid $\delta L/L_0$

Elastic modulus（弹性模量）： $\sigma=Y\varepsilon$

Table 1.2 Comparison of bond types and typical properties (general trends)

Bond Type	Typical Solids	Bond Energy (eV/atom)	Melt. Temp. (°C)	Elastic Modulus (GPa)	Density (g cm^{-3})	Typical Properties
Ionic	NaCl (rock salt)	3.2	801	40	2.17	Generally electrical insulators. May become conductive at high temperatures.
	MgO (magnesia)	10	2852	250	3.58	High elastic modulus. Hard and brittle but cleavable. Thermal conductivity less than metals.
Metallic	Cu	3.1	1083	120	8.96	Electrical conductor.
	Mg	1.1	650	44	1.74	Good thermal conduction. High elastic modulus. Generally ductile. Can be shaped.
Covalent	Si	4	1410	190	2.33	Large elastic modulus. Hard and brittle.
	C (diamond)	7.4	3550	827	3.52	Diamond is the hardest material. Good electrical insulator. Moderate thermal conduction, though diamond has exceptionally high thermal conductivity.
van der Waals: hydrogen bonding	PVC (polymer)		212	4	1.3	Low elastic modulus. Some ductility.
	H_2O (ice)	0.52	0	9.1	0.917	Electrical insulator. Poor thermal conductivity. Large thermal expansion coefficient.
van der Waals: induced dipole	Crystalline argon	0.09	−189	8	1.8	Low elastic modulus. Electrical insulator. Poor thermal conductivity. Large thermal expansion coefficient.

The higher the bond energy, the higher the melting point, the greater the modulus of elasticity, and the smaller the coefficient of thermal expansion.